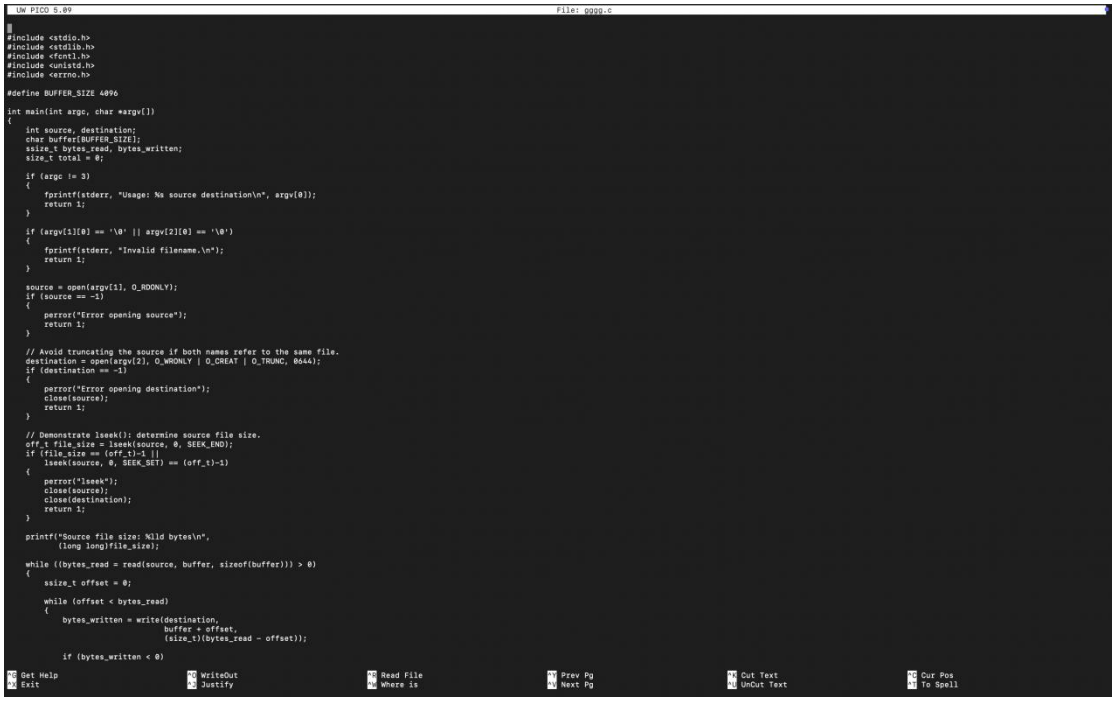
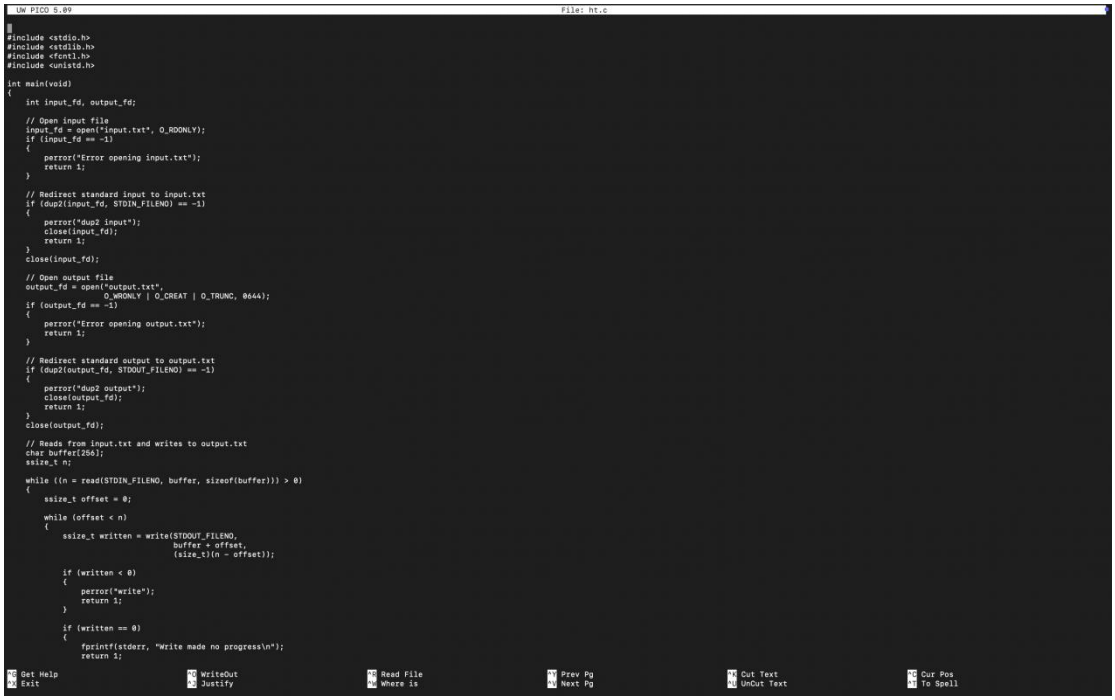
NAME: B SREE NIKHIL REDDY ID NO: 2520090058 PRACTICAL WEEK 9

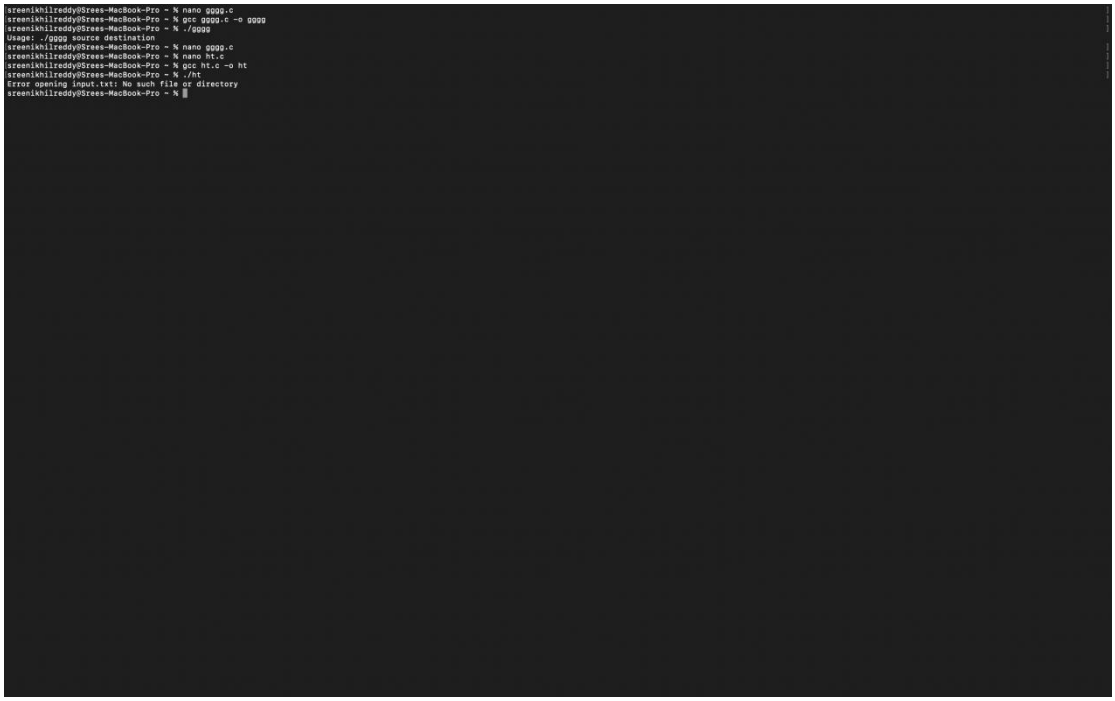
1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite)





2. Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection


UW PICO 5.09
File: ht.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
int main(void)
{
int input_fd, output_fd;
// Open input file
input_fd = open("input.txt", O_RDONLY);
if (input_fd == -1)
{
perror("Error opening input.txt");
return 1;
}
// Redirect standard input to input.txt
if (dup2(input_fd, STDIN_FILENO) == -1)
{
perror("dup2 input");
close(input_fd);
return 1;
}
close(input_fd);
// Open output file
output_fd = open("output.txt",
O_WRONLY | O_CREAT | O_TRUNC, 0644);
if (output_fd == -1)
{
perror("Error opening output.txt");
return 1;
}
// Redirect standard output to output.txt
if (dup2(output_fd, STDOUT_FILENO) == -1)
{
perror("dup2 output");
close(output_fd);
return 1;
}
close(output_fd);
// Reads from input.txt and writes to output.txt
char buffer[256];
ssize_t n;
while ((n = read(STDIN_FILENO, buffer, sizeof(buffer))) > 0)
{
ssize_t offset = 0;
while (offset < n)
{
ssize_t written = write(STDOUT_FILENO,
buffer + offset,
(size_t)(n - offset));
if (written < 0)
{
perror("write");
return 1;
}
if (written == 0)
{
fprintf(stderr, "Write made no progress\n");
return 1;
^G Get Help
^X Exit
^O WriteOut
^J Justify
^R Read File
^W Where is
^Y Prev Pg
^V Next Pg
^K Cut Text
^U UnCut Text
^C Cur Pos
^T To Spell


sreenikhilreddy@Srees-MacBook-Pro ~ % nano gggg.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc gggg.c -o gggg
sreenikhilreddy@Srees-MacBook-Pro ~ % ./gggg
Usage: ./gggg source destination
sreenikhilreddy@Srees-MacBook-Pro ~ % nano gggg.c
sreenikhilreddy@Srees-MacBook-Pro ~ % nano ht.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc ht.c -o ht
sreenikhilreddy@Srees-MacBook-Pro ~ % ./ht
Error opening input.txt: No such file or directory
sreenikhilreddy@Srees-MacBook-Pro ~ %